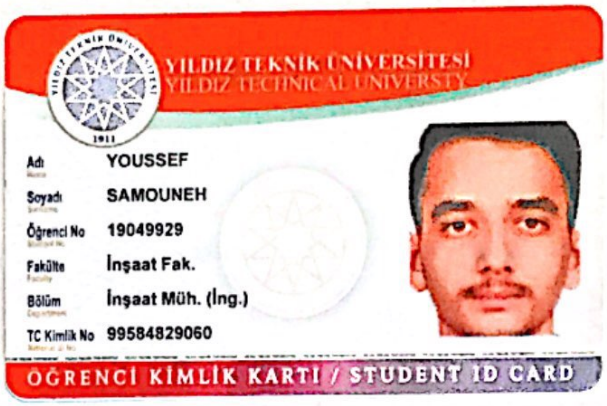
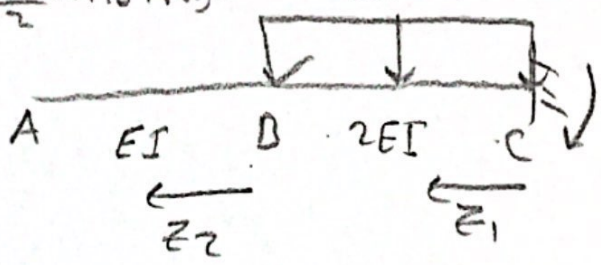
Name: YOUSSEF
Surname: SAMOUNEH
ID No: 19049929
Group: 5



$Q = 20N \qquad L = 192\,cm$

$0 < z_1 < L \qquad q = 20 \qquad M_z = -\frac{q \cdot z^2}{2} - M_0 + R_{cy} z$

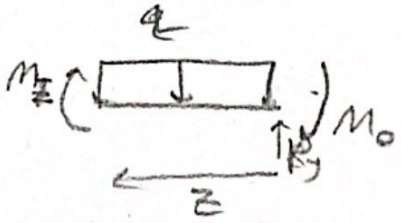
$0 < z_2 < L \qquad q = 0$



$$2EIV'' = \frac{qz^2}{2} + M_0 - R_{cy} z$$

$$2EIV' = \frac{qz^3}{6} + M_0 z - \frac{R_{cy} z^2}{2} + C_1$$

$$2EIV = \frac{qz^4}{24} + \frac{M_0 z^2}{2} - \frac{R_{cy} z^3}{6} + C_1 z + C_2$$



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$(V_{z=0} = 0) \qquad C_2 = 0 \qquad V'_{z=0} = 0 \qquad C_1 = 0$

$\Omega(z) = V'(z) \qquad 2EI\,\Omega(z) = \frac{qz^3}{6} + M_0 z - \frac{R_{cy} z^2}{2}$

$2EI\,V(z) = \frac{qz^4}{24} + \frac{M_0 z^2}{2} - \frac{R_{cy} z^3}{6}$

At (A)

$\Omega(z = 2L) = 0 \qquad 0 = \frac{20(2 \cdot 192)^2}{6} + M_0 (2 \cdot 192) - \frac{R_{cy} (2 \cdot 192)^2}{2}$

$M_0 = \frac{73728 R_{cy} -}{2 \cdot 192} \qquad = 1908 R_{cy} - 149520 = M_0$

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